Introduction to Digital Systems

Digital System Fundamentals

Sorayut Glomglome

Objectives

- Explain the basic differences between digital and analog quantities
- Show how voltage levels are used to represent digital quantities
- Explain the basic logic functions of NOT, AND, and OR
- Describe several types of logic operations

Contents

1. Digital and Analog Quantities
2. Binary Digits, Logic Levels, and Digital Waveforms
3. Basic Logic Functions
4. Combinational and Sequential Logic Functions
5. Fixed-Function Logic Devices

1. Digital and Analog Quantities

Analog Quantities

- Most natural quantities that we see are analog and vary continuously. Analog systems can generally handle higher power than digital systems.

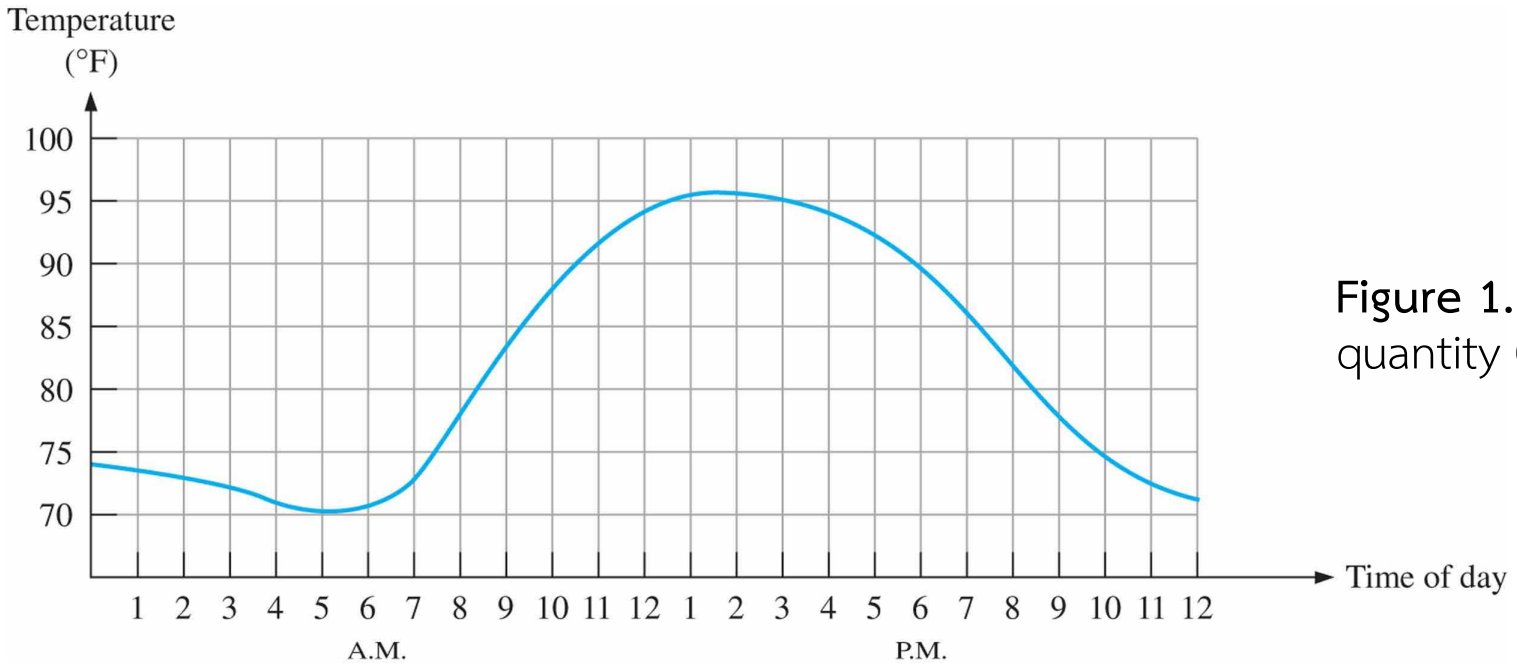
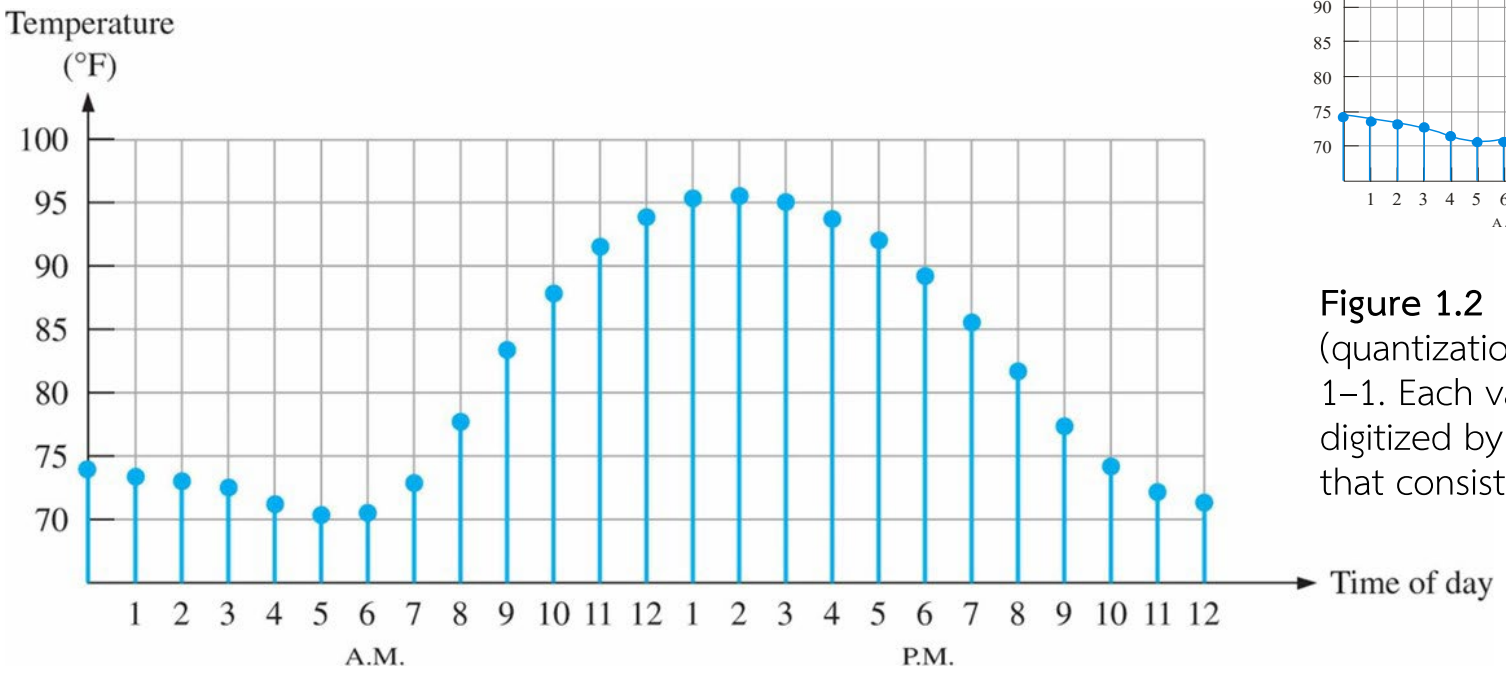


Figure 1.1 Graph of an analog quantity (temperature versus time).

Digital Advantage

- A digital quantity is one having a discrete set of values.



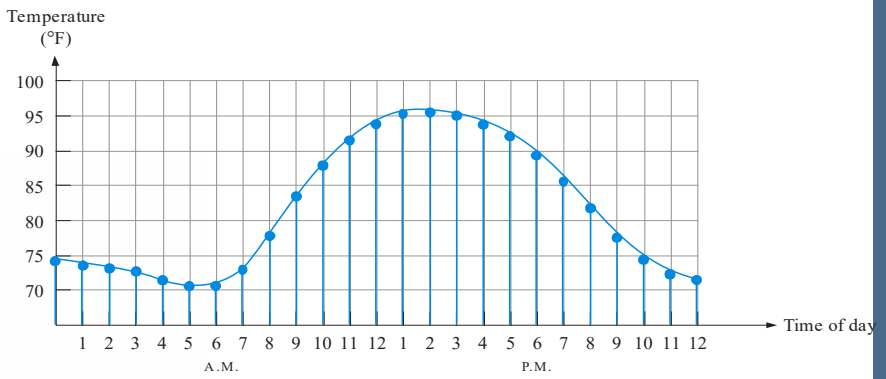


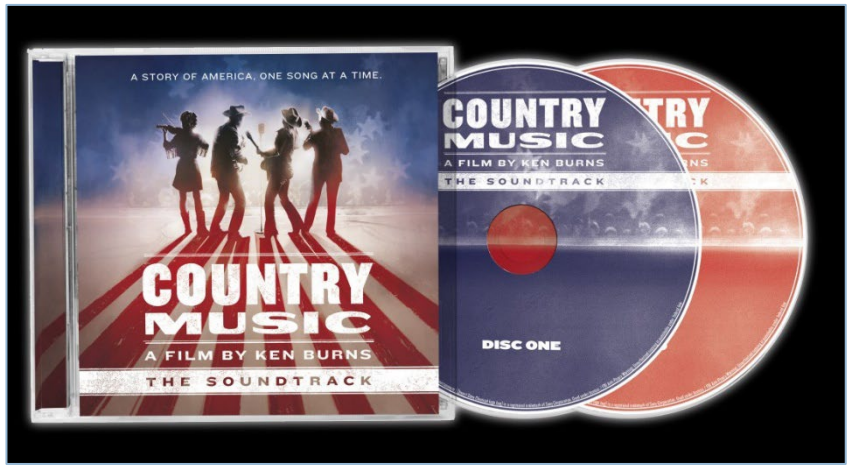
Figure 1.2 Sampled-value representation (quantization) of the analog quantity in Figure 1–1. Each value represented by a dot can be digitized by representing it as a digital code that consists of a series of 1s and 0s.

Digital Advantage

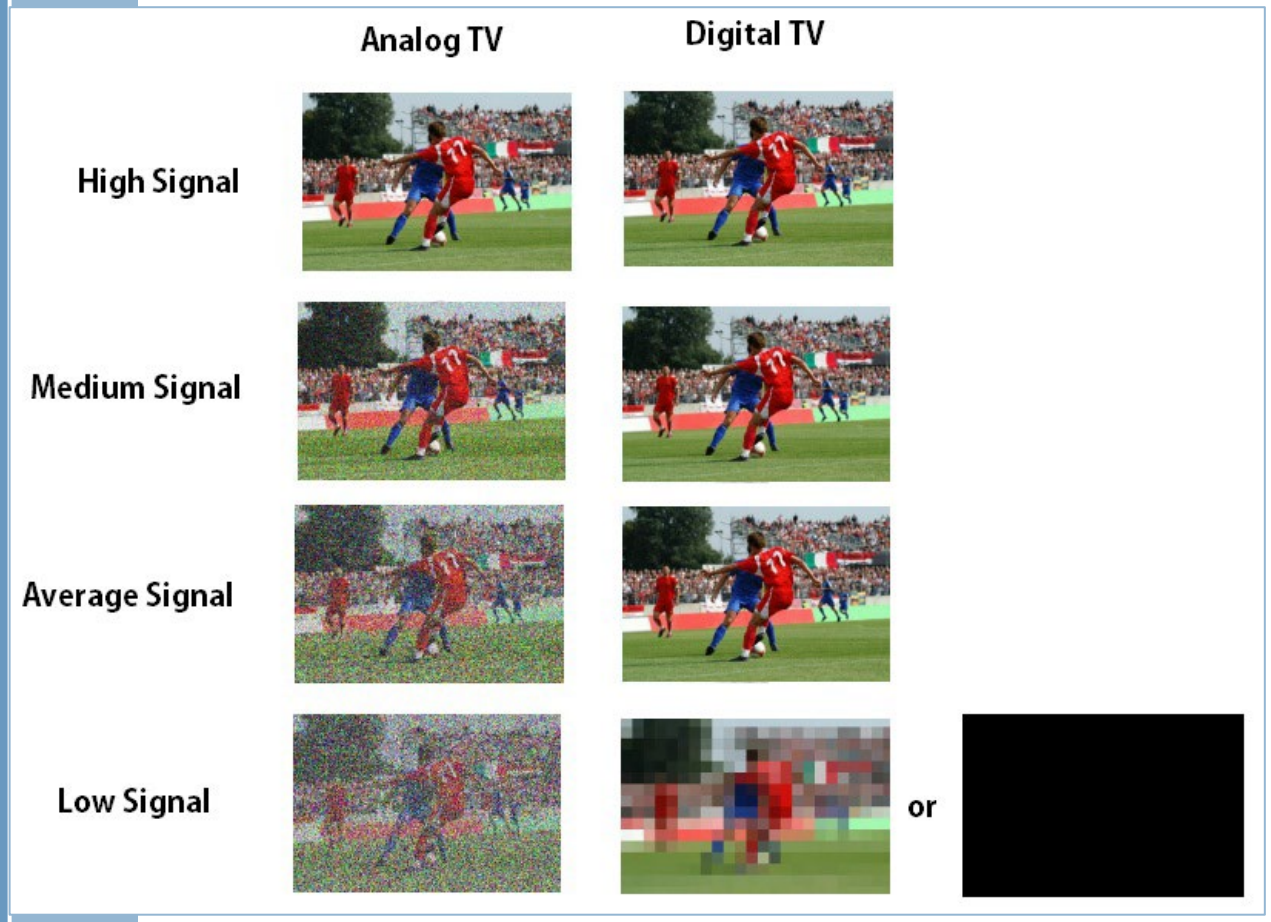
- Processed and transmitted more efficiently and reliably
- Storage -> need, compact, reproduced quality

Digital Advantage

- Music CD vs Vinyl
  - Reproduction quality
  - Clarity
  - Compact (Data size)



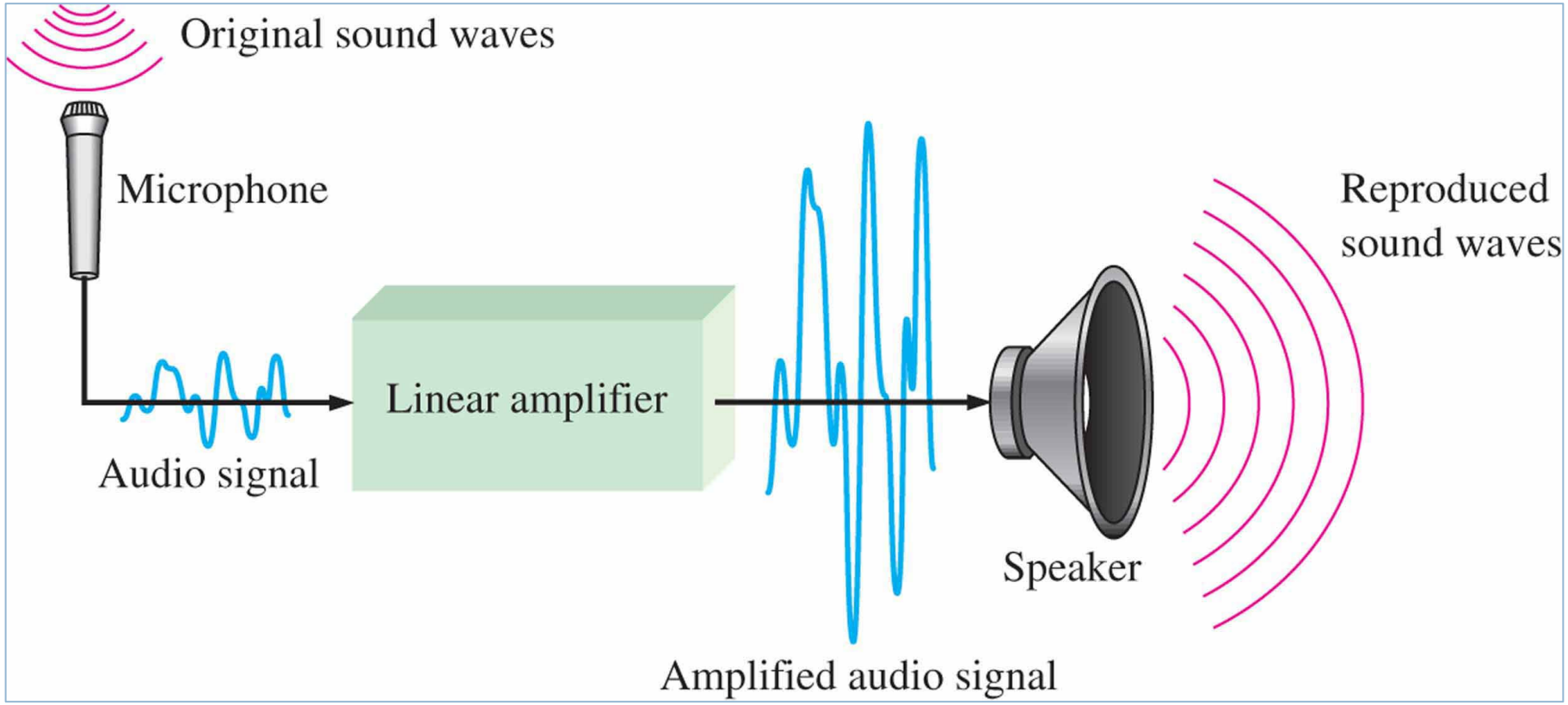
Digital Advantage



- Analog TV vs Digital TV
  - Noise

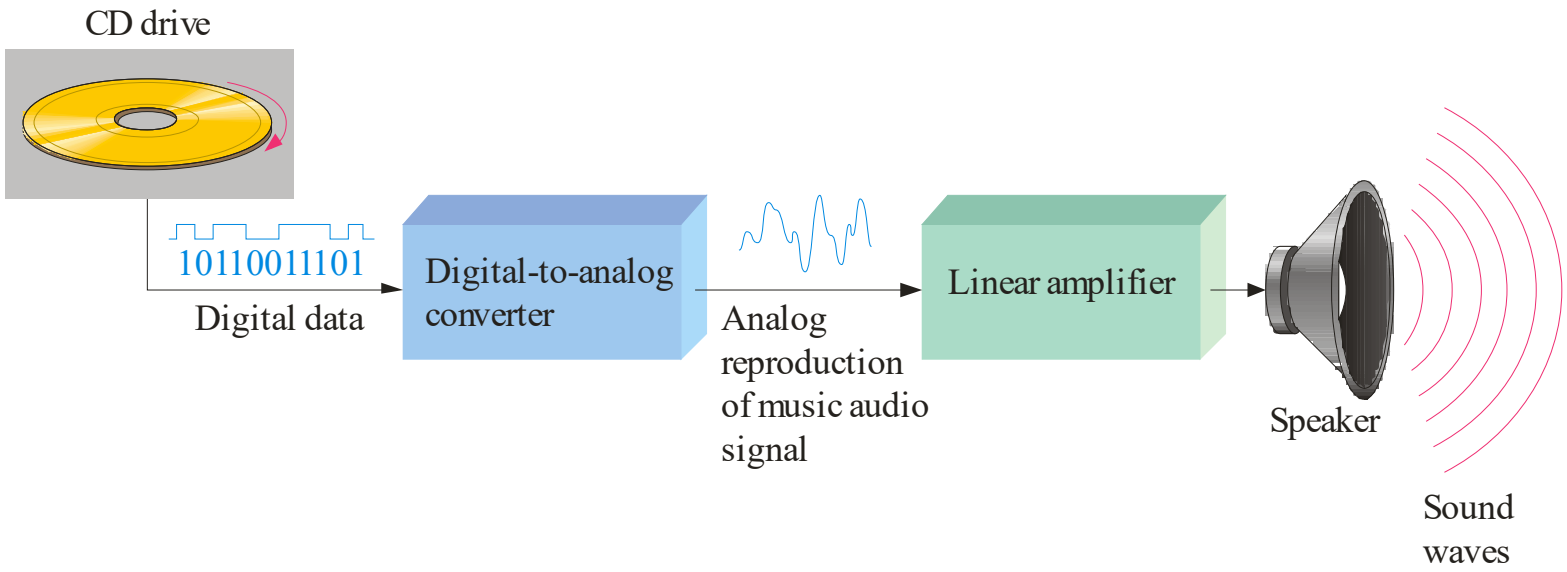
Digital TV Information - Antenna Direct

Analog System



Mixed Analog and Digital System

- Many systems use a mix of analog and digital electronics to take advantage of each technology.
- CD player accepts digital data from the CD drive and converts it to an analog signal for amplification.

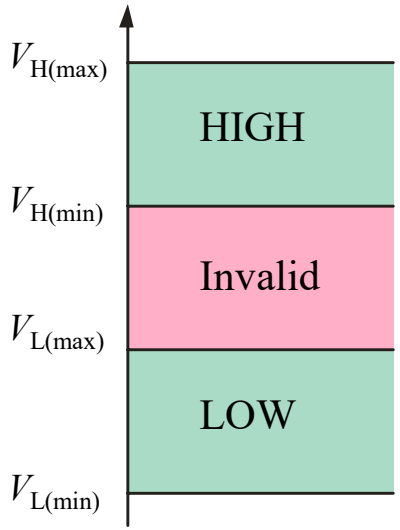


2. Binary Digits, Logic Levels, and Digital Waveforms

Binary Digits and Logic Levels

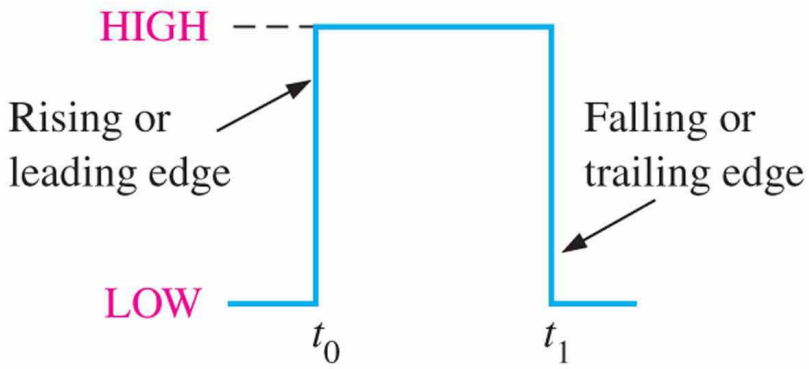
- Digital electronics uses circuits that have two states, which are represented by two different voltage levels called HIGH and LOW. The voltages represent numbers in the binary system.

- In binary, a single number is called a bit (for *b*inary dig*it*).
- A bit can have the value of either a 0 or a 1, depending on if the voltage is LOW or HIGH.

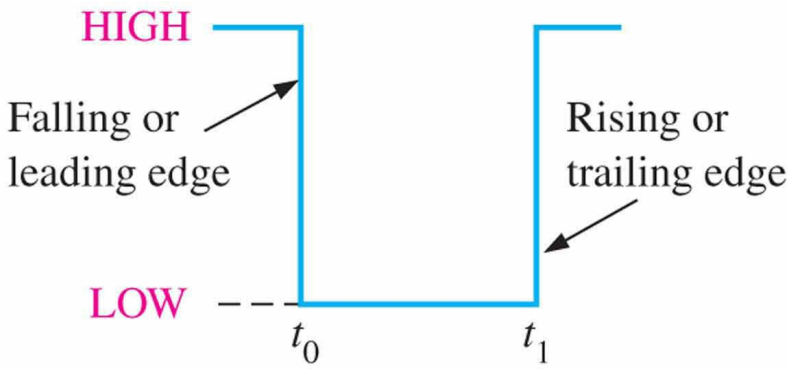


Digital Waveforms

- Digital waveforms change between the LOW and HIGH levels.
- A positive going pulse is one that goes from a normally LOW logic level to a HIGH level and then back again.
- Digital waveforms are made up of a series of pulses



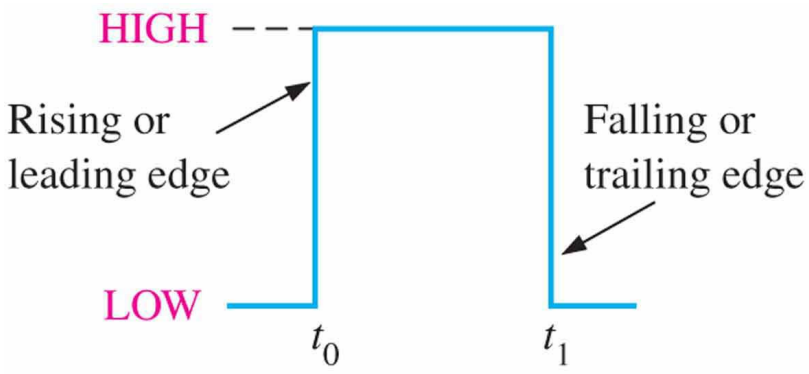
(a) Positive–going pulse

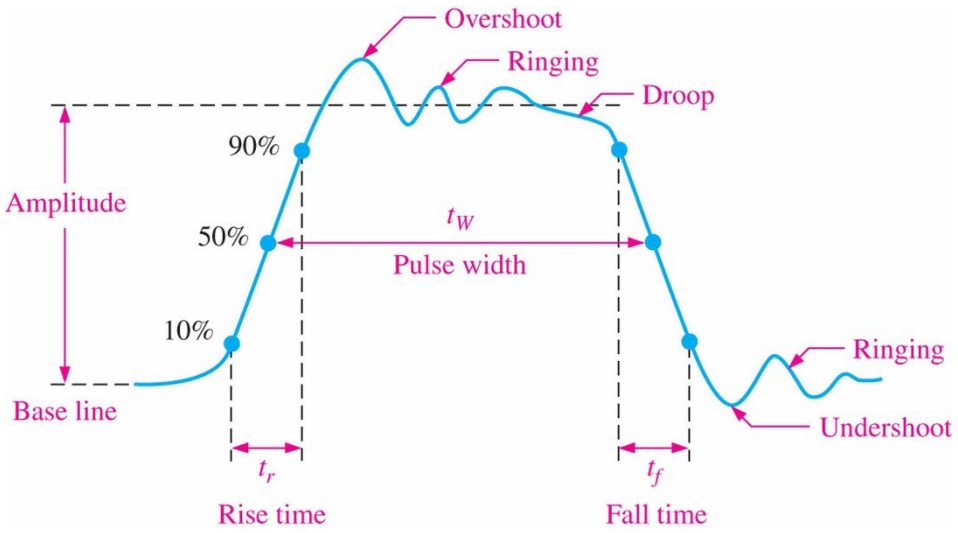


(b) Negative–going pulse

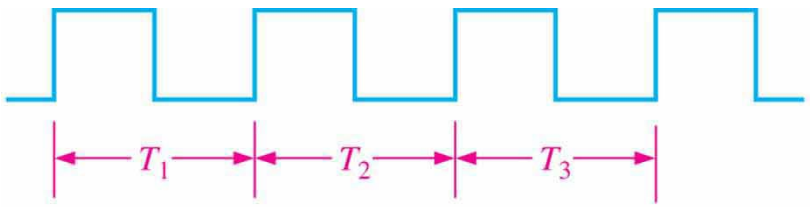
Pulse Definitions

- Actual pulses are not ideal.
- Described by the **rise time**, **fall time**, **amplitude**, and **other characteristics**.





Periodic & Non Periodic Waveforms



Period $= T_1 = T_2 = T_3 = \ldots = T_n$

Frequency $= \frac{1}{T}$

(a) Periodic (square wave)

(b) Nonperiodic

The frequency (f) of a pulse (digital) waveform is the **reciprocal** of the period. The relationship between frequency and period is expressed as follows:

$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

Duty Cycle

An important characteristic of a periodic digital waveform is its **duty cycle**, which is the ratio of the pulse width (t_W) to the period (T). It can be expressed as a percentage.

$$\text{Duty cycle} = \left(\frac{t_W}{T}\right)100\% \qquad \text{Equation 1–3}$$

EXAMPLE 1–1

A portion of a periodic digital waveform is shown in Figure 1–10. The measurements are in milliseconds. Determine the following:

(a) period (b) frequency (c) duty cycle

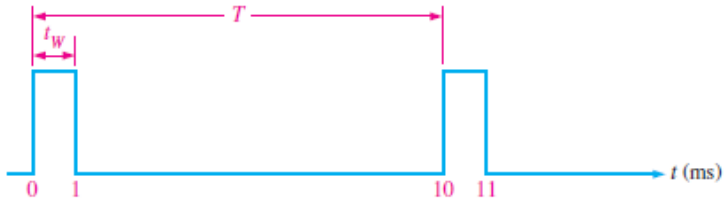


FIGURE 1–10

Solution

(a) The period (T) is measured from the edge of one pulse to the corresponding edge of the next pulse. In this case T is measured from leading edge to leading edge, as indicated. T equals **10 ms**.

(b) $f = \frac{1}{T} = \frac{1}{10\text{ ms}} = \mathbf{100\ Hz}$

(c) $\text{Duty cycle} = \left(\frac{t_W}{T}\right)100\% = \left(\frac{1\text{ ms}}{10\text{ ms}}\right)100\% = \mathbf{10\%}$

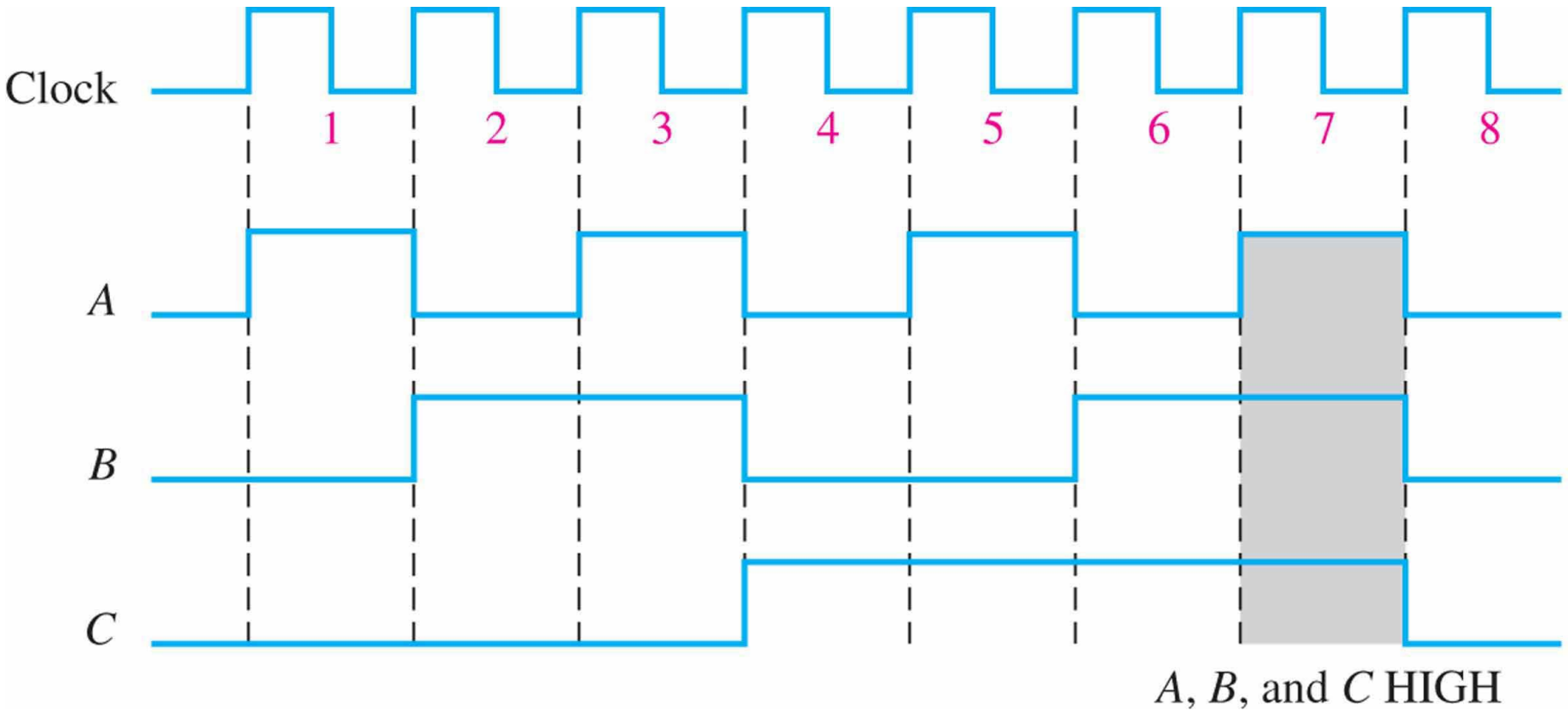
Related Problem*

A periodic digital waveform has a pulse width of 25 μs and a period of 150 μs. Determine the frequency and the duty cycle.

*Answers are at the end of the chapter.

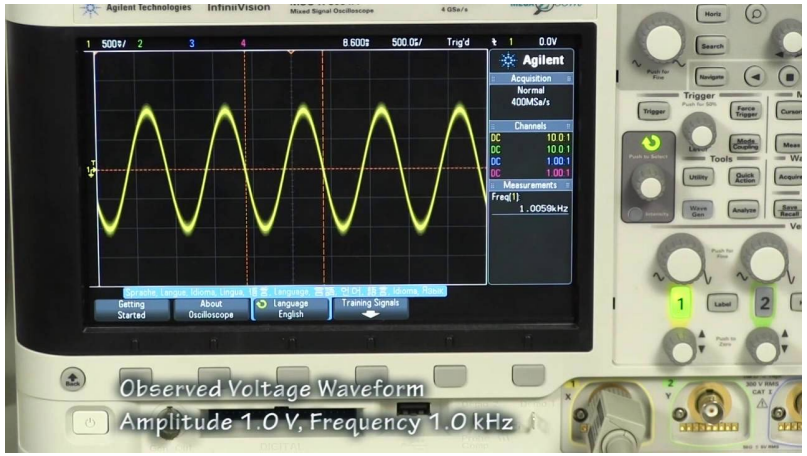
Timing Diagram

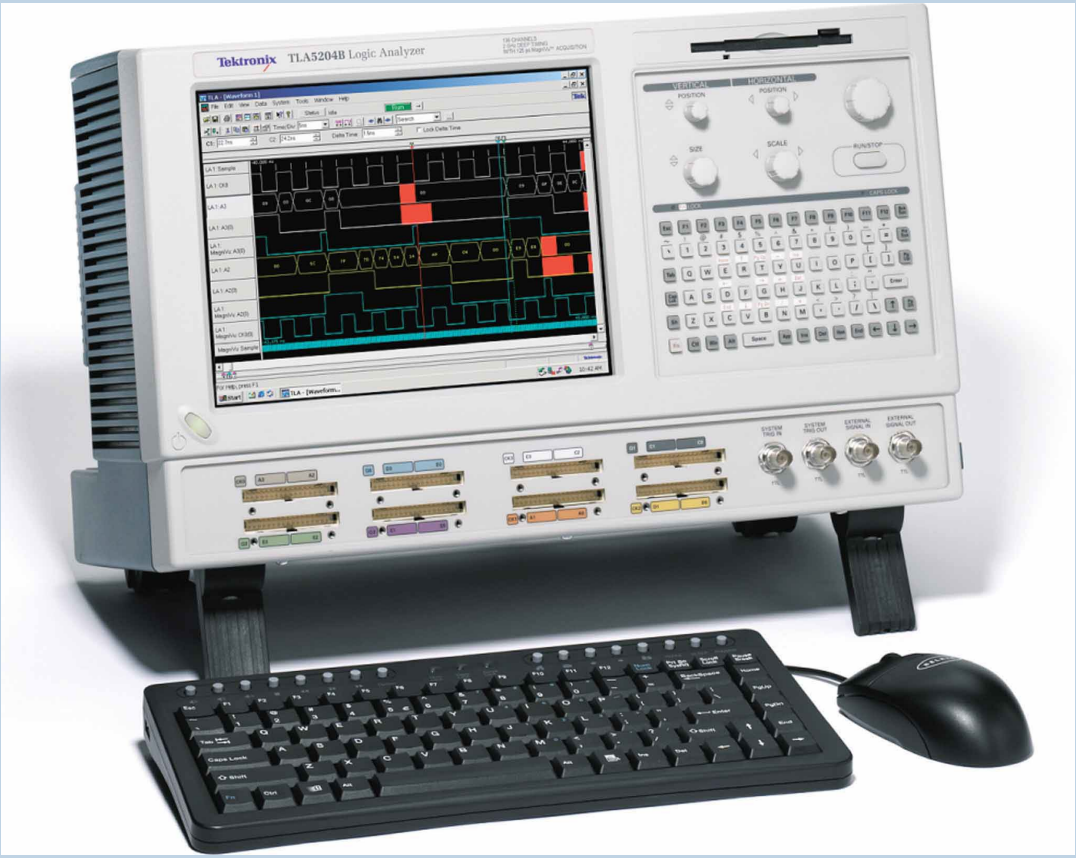
- A timing diagram is used to show the relationship between two or more digital waveforms,



LOGIC ANALYZER

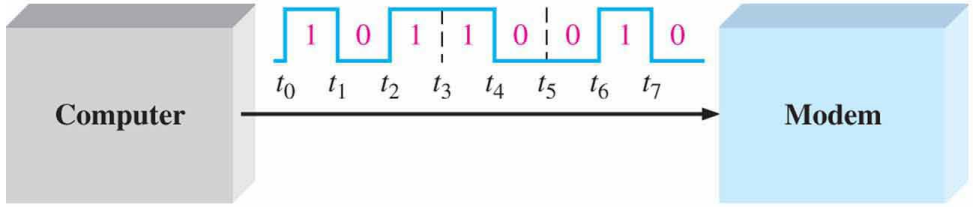
Observes timing diagram



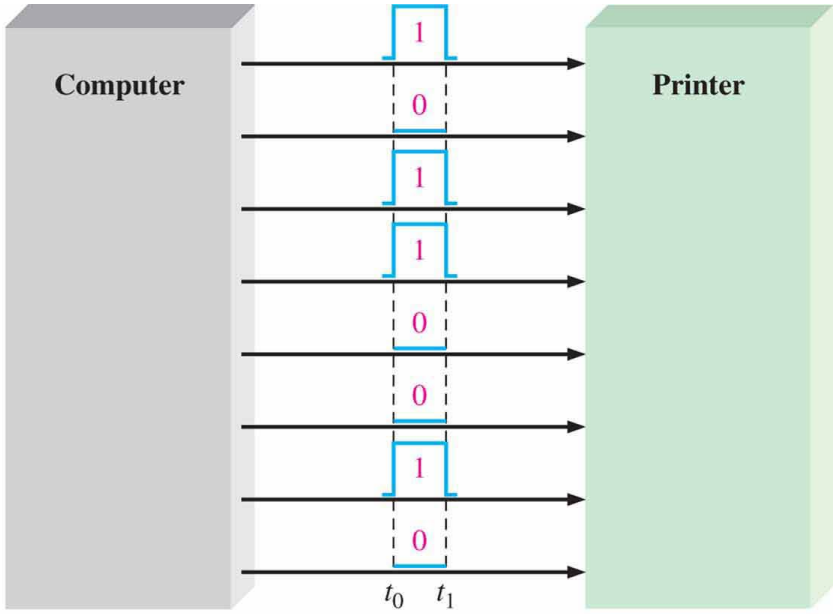


Serial and Parallel Data Transfer

- Data can be transmitted by either serial transfer or parallel transfer.



(a) Serial transfer of 8 bits of binary data from computer to modem. Interval t_0 to t_1 is first.



(b) Parallel transfer of 8 bits of binary data from computer to printer. The beginning time is t_0.

Data Transfer

EXAMPLE 1–2

(a) Determine the total time required to serially transfer the eight bits contained in waveform *A* of Figure 1–14, and indicate the sequence of bits. The left-most bit is the first to be transferred. The 1 MHz clock is used as reference.

(b) What is the total time to transfer the same eight bits in parallel?

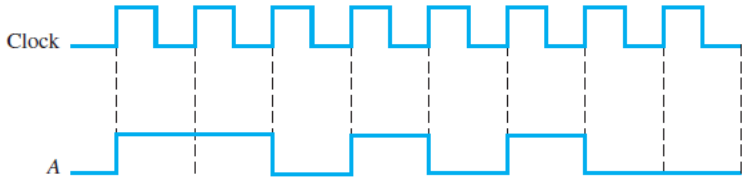


FIGURE 1–14

Solution

(a) Since the frequency of the clock is 1 MHz, the period is

$$T = \frac{1}{f} = \frac{1}{1\ \text{MHz}} = 1\ \mu\text{s}$$

It takes 1 μs to transfer each bit in the waveform. The total transfer time for 8 bits is

$$8 \times 1\ \mu\text{s} = \mathbf{8\ \mu s}$$

To determine the sequence of bits, examine the waveform in Figure 1–14 during each bit time. If waveform *A* is HIGH during the bit time, a 1 is transferred. If waveform *A* is LOW during the bit time, a 0 is transferred. The bit sequence is illustrated in Figure 1–15. The left-most bit is the first to be transferred.

FIGURE 1–15

(b) A parallel transfer would take **1 μs** for all eight bits.

Related Problem

If binary data are transferred on a USB at the rate of 480 million bits per second (480 Mbps), how long will it take to serially transfer 16 bits?

3. Basic Logic Functions

Logic

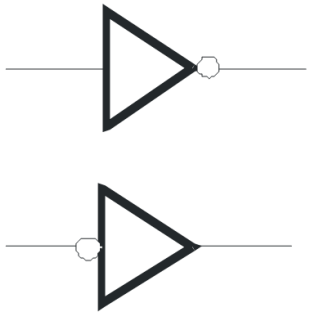
- Two-state characteristics
  - ON-OFF, 0-1, LOW-HIGH, YES-NO or TRUE-FALSE
- Logic is applied to digital circuits used to implement logic functions
  - Building blocks for such complex digital systems as the computer

Basic Logic Functions

AND True only if ***all*** input conditions are true.

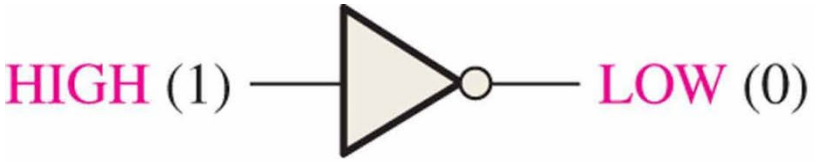
OR True only if ***one or more*** input conditions are true.

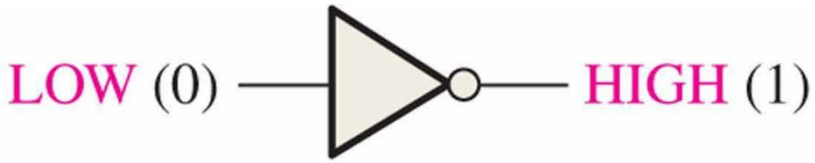
NOT Indicates the ***opposite*** condition.



NOT Function / NOT Gate

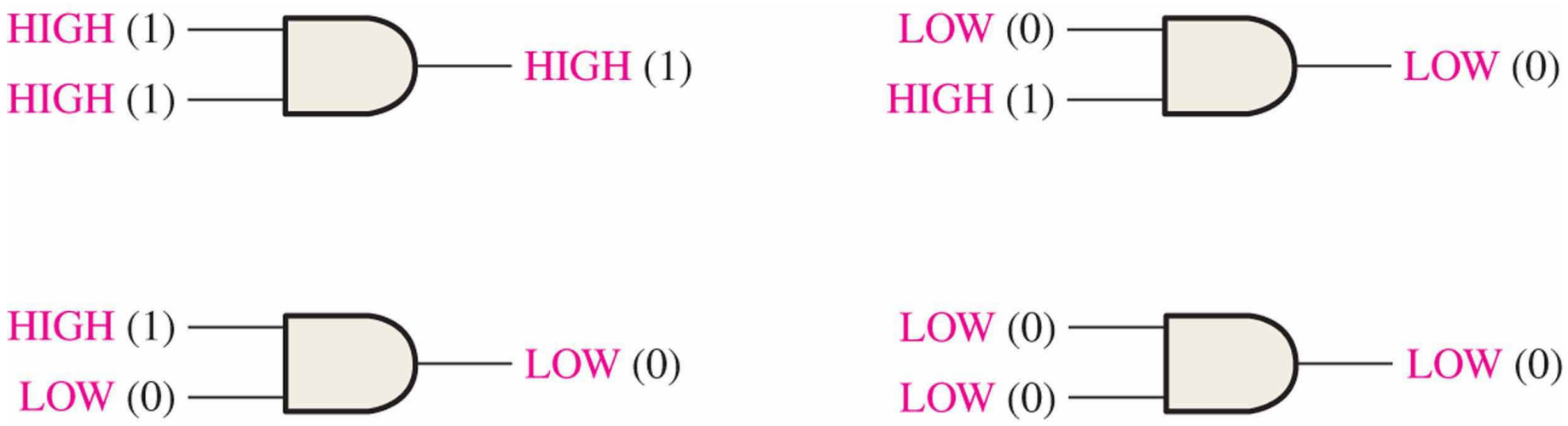
- Changes one logic level to the opposite logic level
- aka Inverter





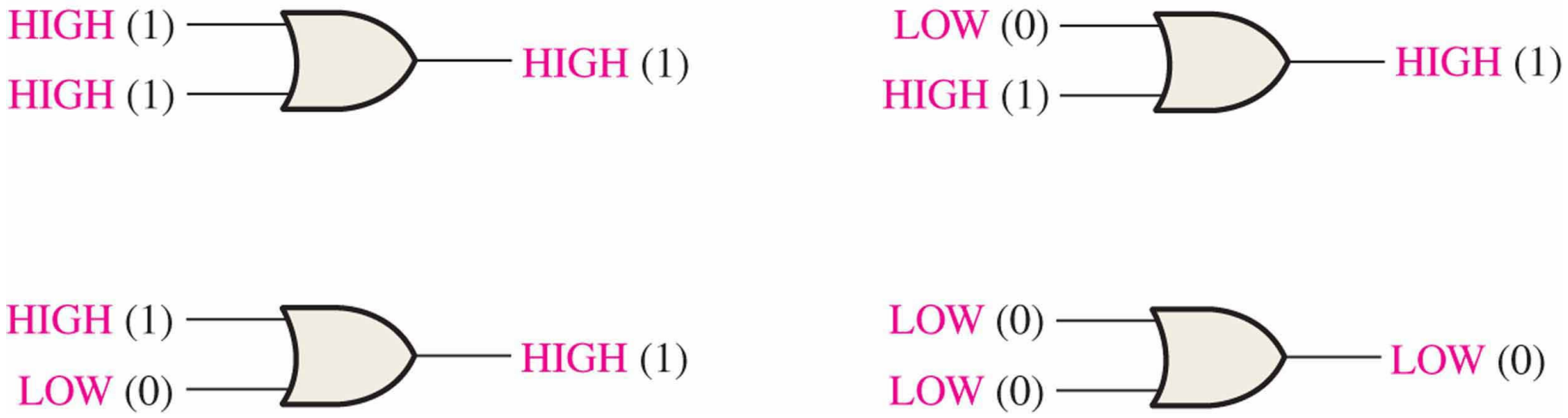
AND Function / AND Gate

- Produces a HIGH output only when all the inputs are HIGH.



OR Function / OR Gate

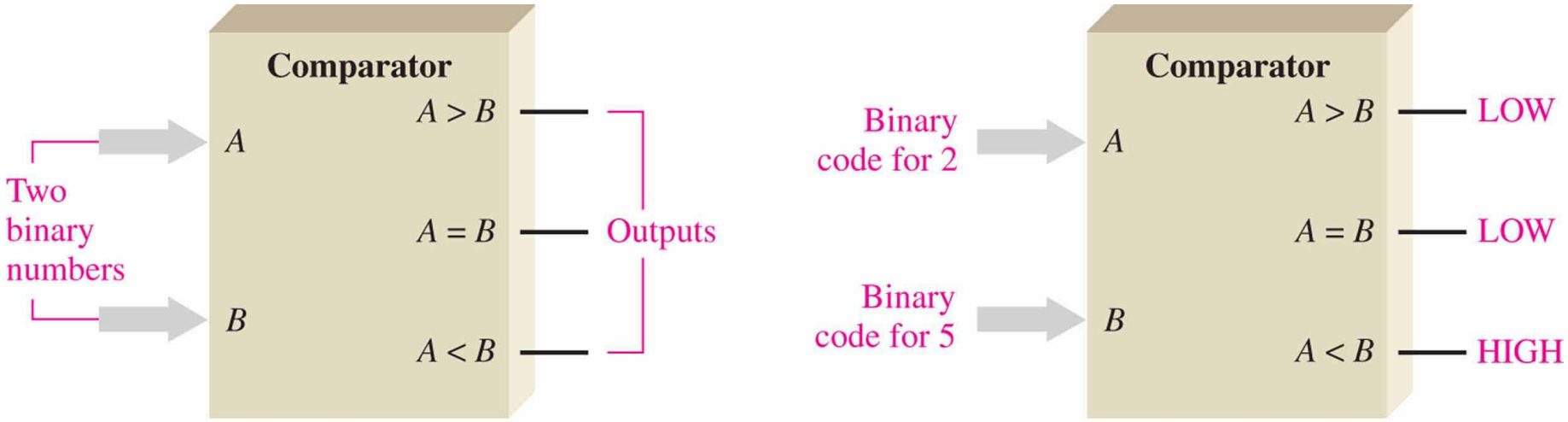
- Produces a HIGH output when one or more inputs are HIGH



4. Combinational and Sequential Logic Functions

Comparator Function/Circuit

- Compares two quantities.

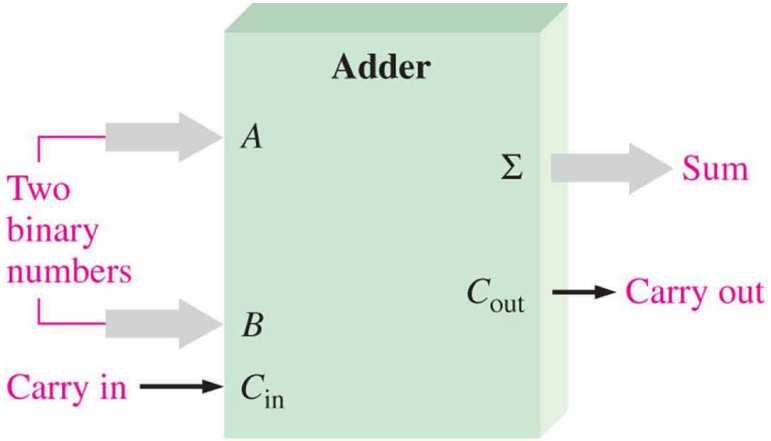


(a) Basic magnitude comparator

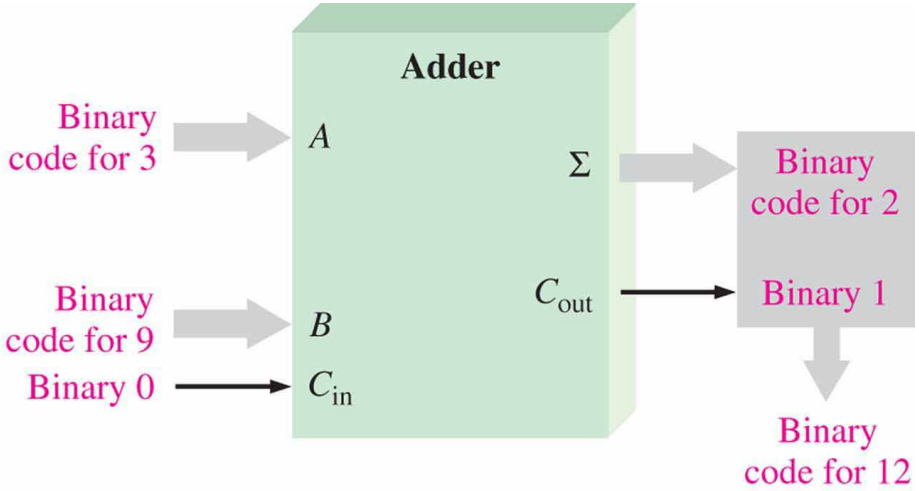
(b) Example: A is less than B ($2 < 5$) as indicated by the HIGH output ($A < B$)

Addition Function / Adder

- Adds two binary numbers and generates a SUM and a CARRY output



(a) Basic adder



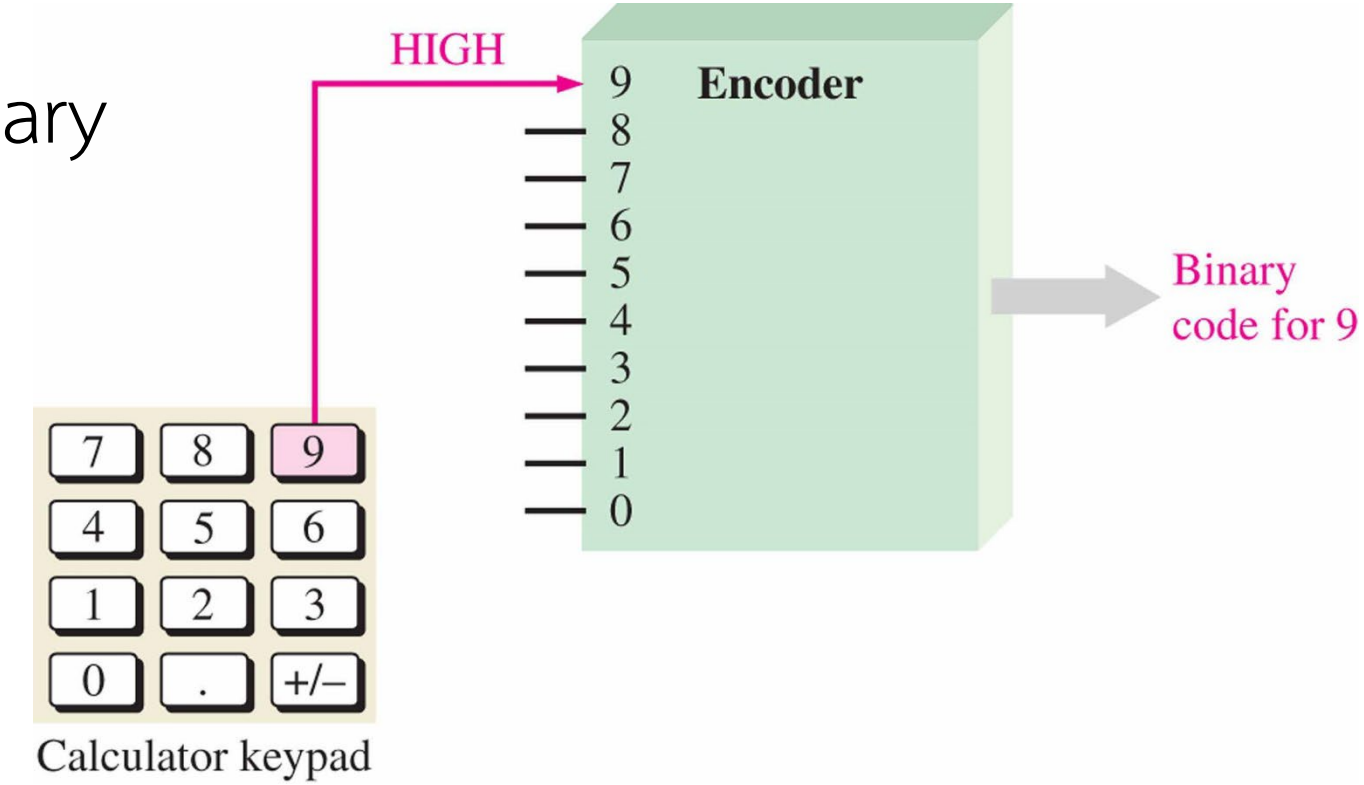
(b) Example: A plus B (3 + 9 = 12)

Other Arithmetic Functions

- Subtraction Function / Subtractor
- Multiplication Function / Multiplier
- Division Function / Divider
  - Series of subtractors

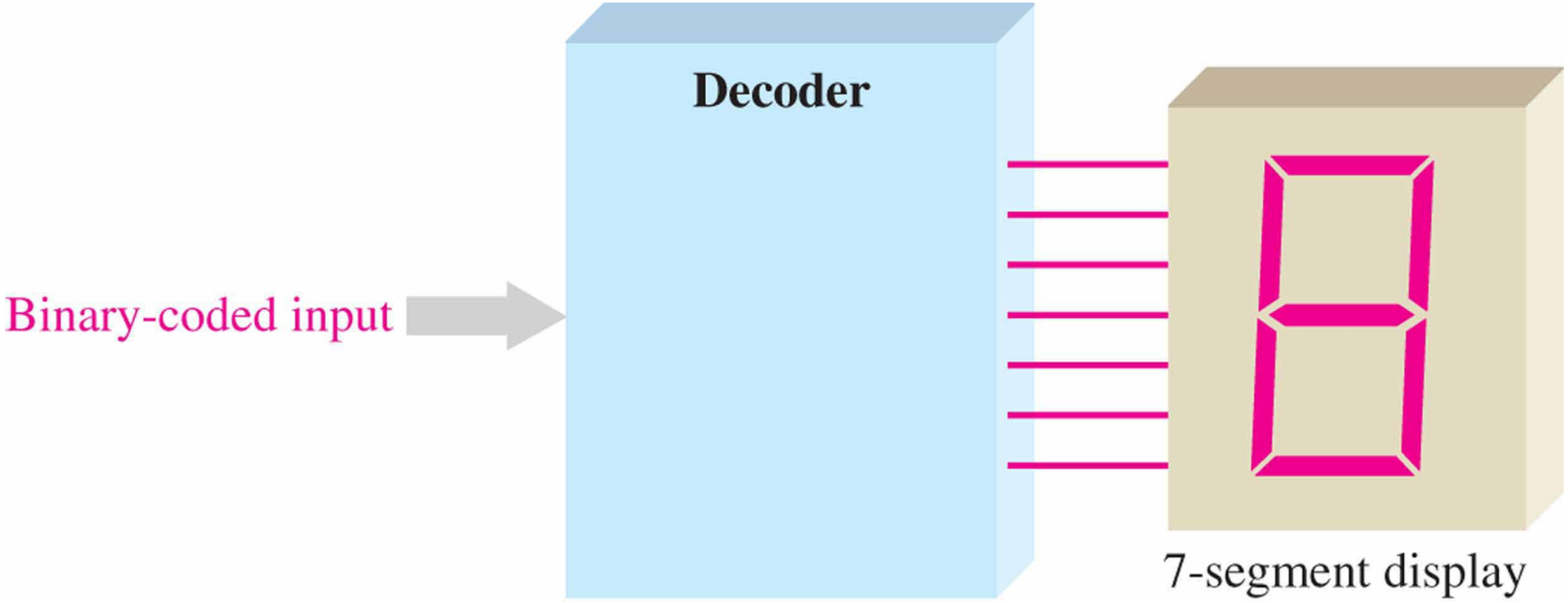
Encoding Function / Encoder

- Convert information
  - Decimal 0-9 to Binary



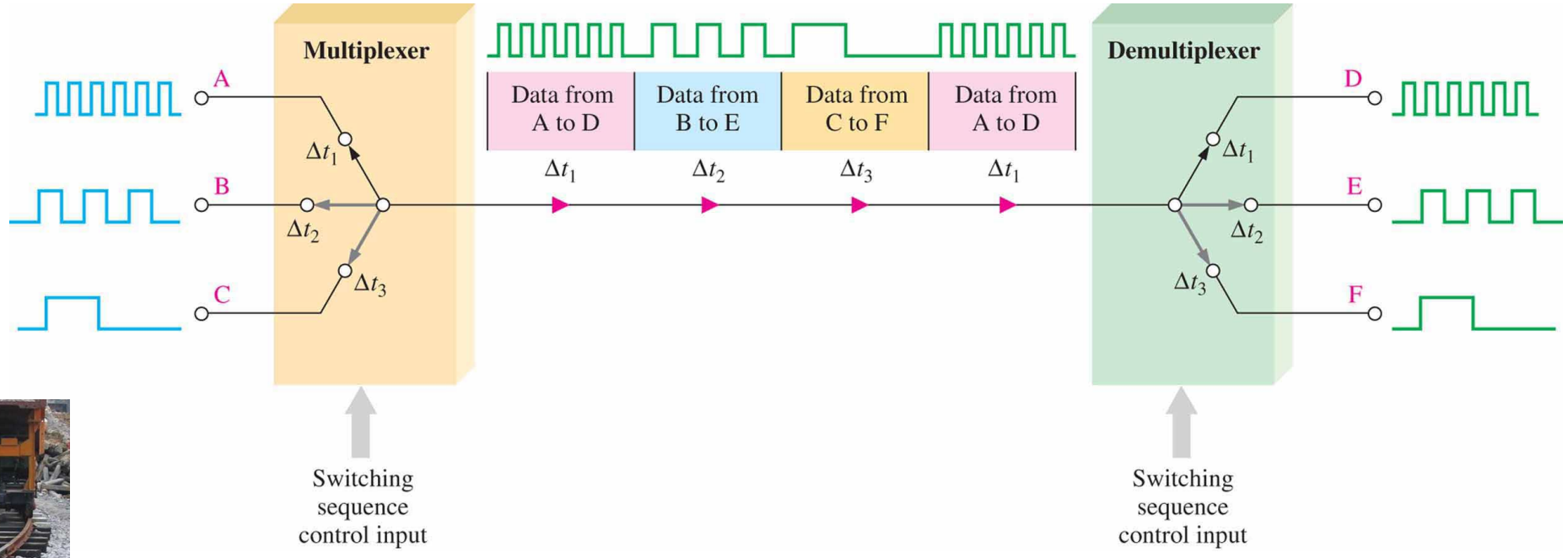
Decoding Function / Decoder

- Converts coded information



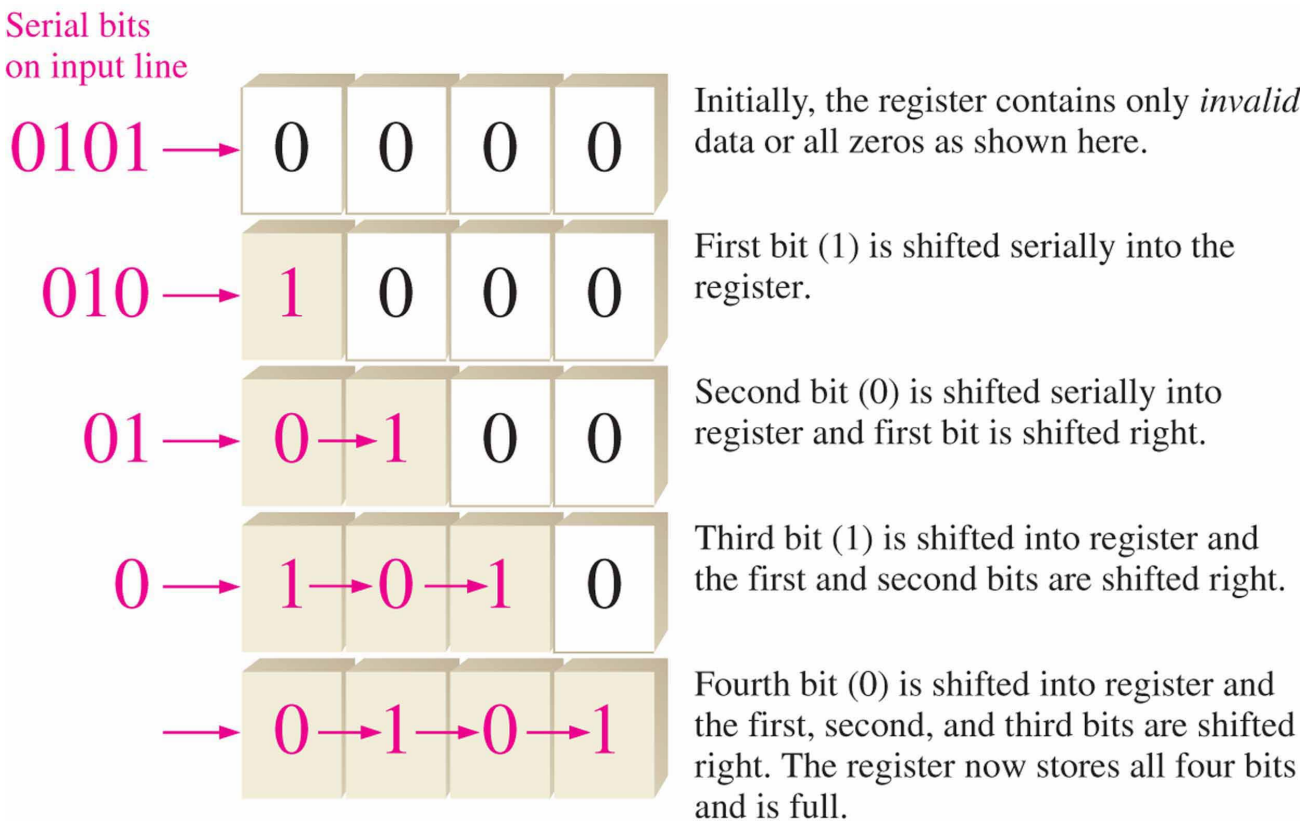
Data Selection Function / Multiplexer (MUX)

- Works with Demultiplexer (DEMUX)

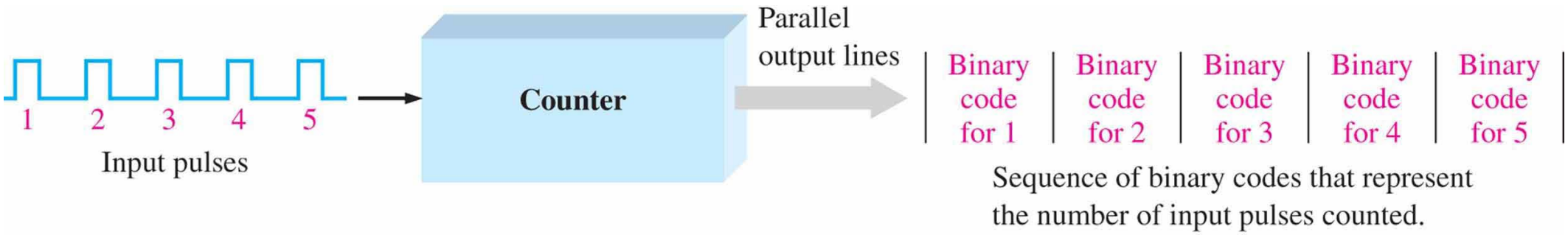


Shift Register and Flip-flops

- A flip-flop can store one bit at a time.
- Shift register moves and stores data each time it is clocked



Counter

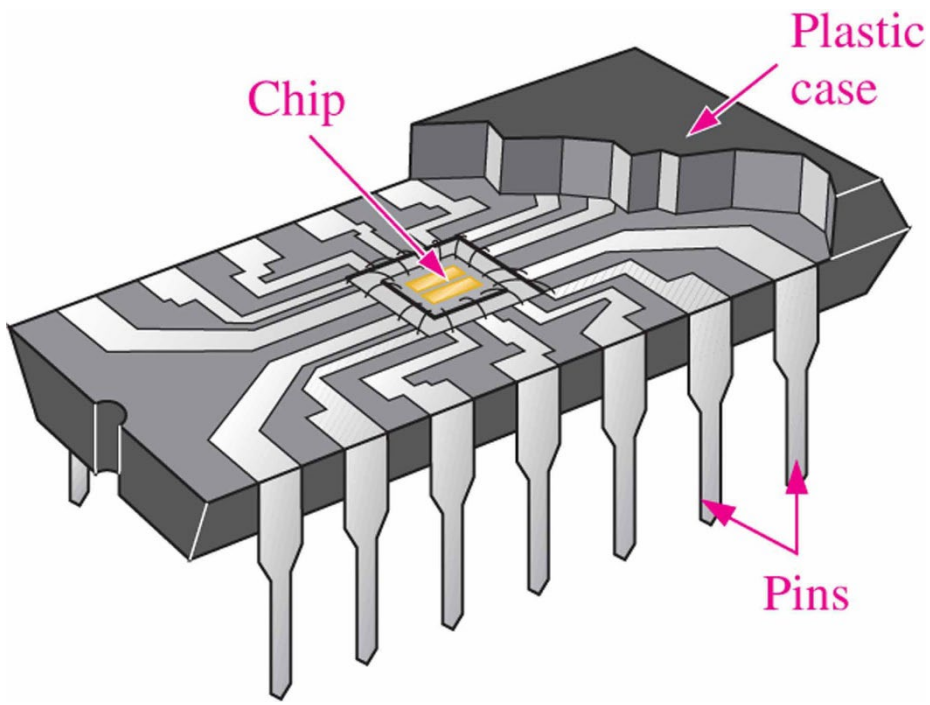

1
2
3
4
5
Input pulses
Counter
Parallel
output lines
Binary
code
for 1
Binary
code
for 2
Binary
code
for 3
Binary
code
for 4
Binary
code
for 5
Sequence of binary codes that represent
the number of input pulses counted.

5. Fixed-Function Logic Devices

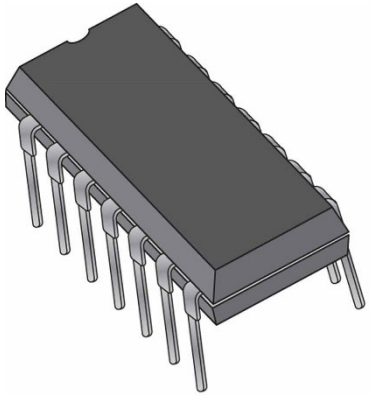
Integrated Circuit

Integrated Circuit (IC)

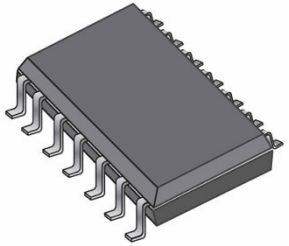
- An electronic circuit constructed on a single small chip of silicon.
  - transistors, diodes, resistors, and capacitors
- Fixed-function logic devices



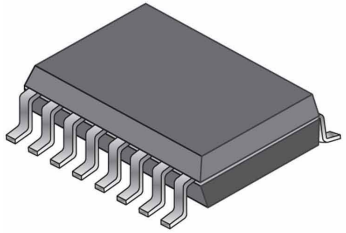
IC Packages



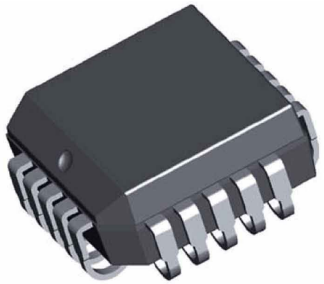
(a) Dual in-line package (DIP)



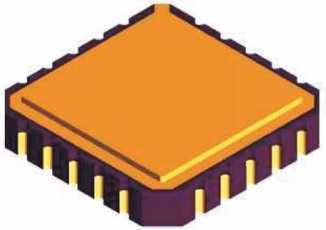
(b) Small-outline IC (SOIC)



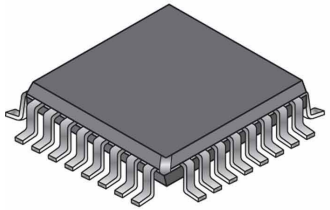
(a) SSOP (153 × 193 mils)



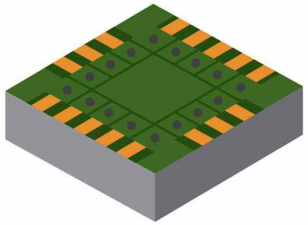
(b) PLCC (350 × 350 mils)



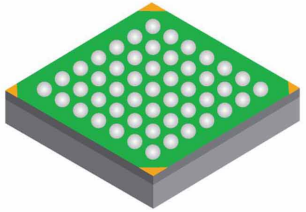
(c) LCC (350 × 350 mils)



(d) LQFP (7 × 7 mm)

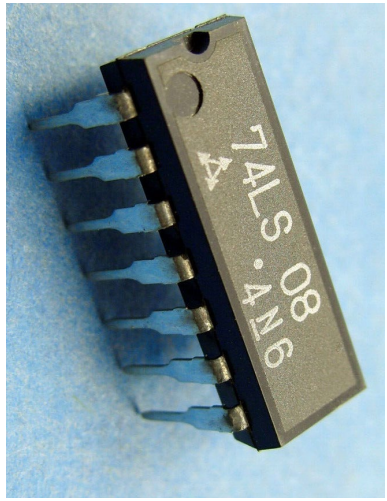


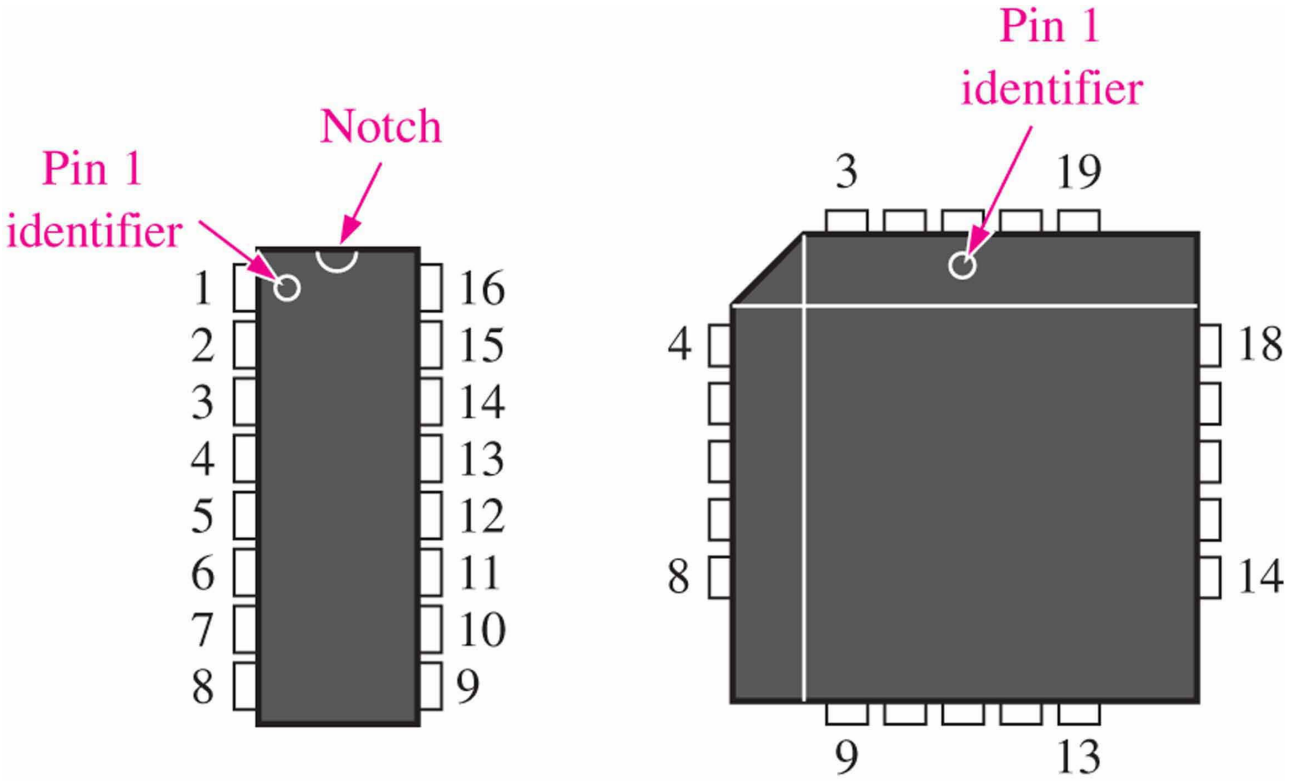
(e) Laminate CSP (3.5 × 3.5 mm)



(f) FBGA (4 × 4 mm)

Pin Numbering

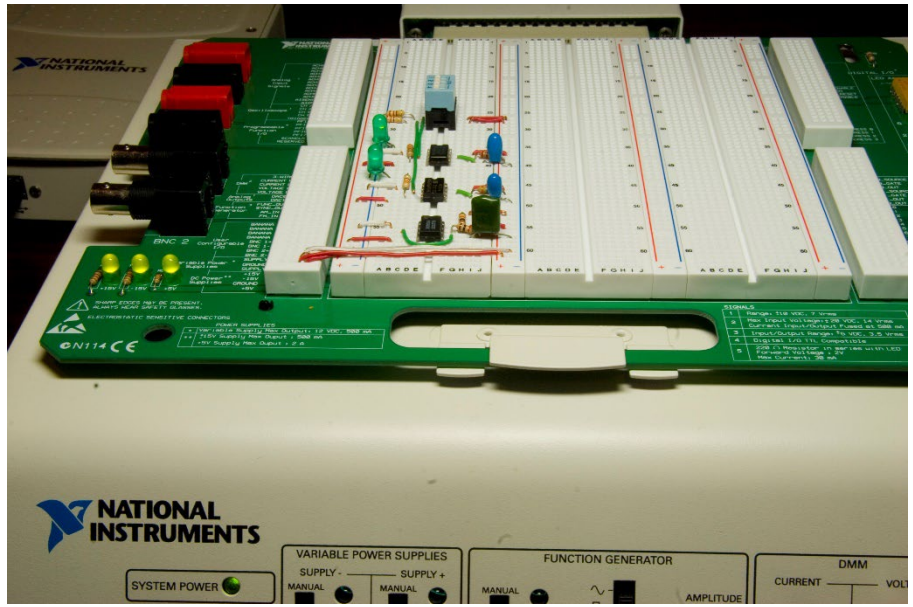


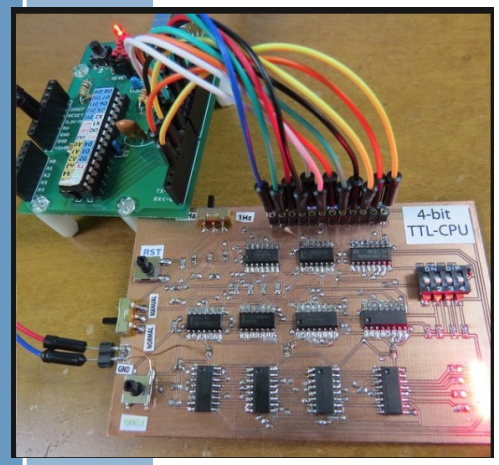


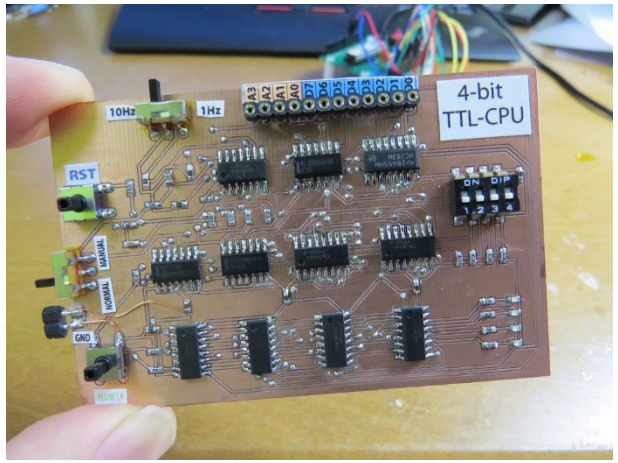
(a) DIP or SSOP (b) PLCC or LCC

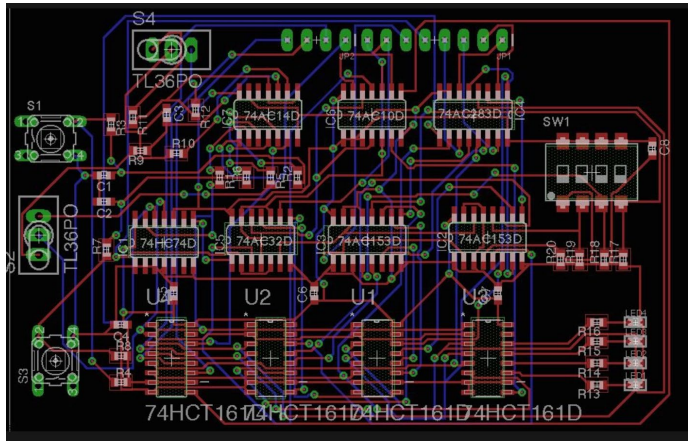
IC Usage

- Circuit built from DIP chips ready to be tesed.









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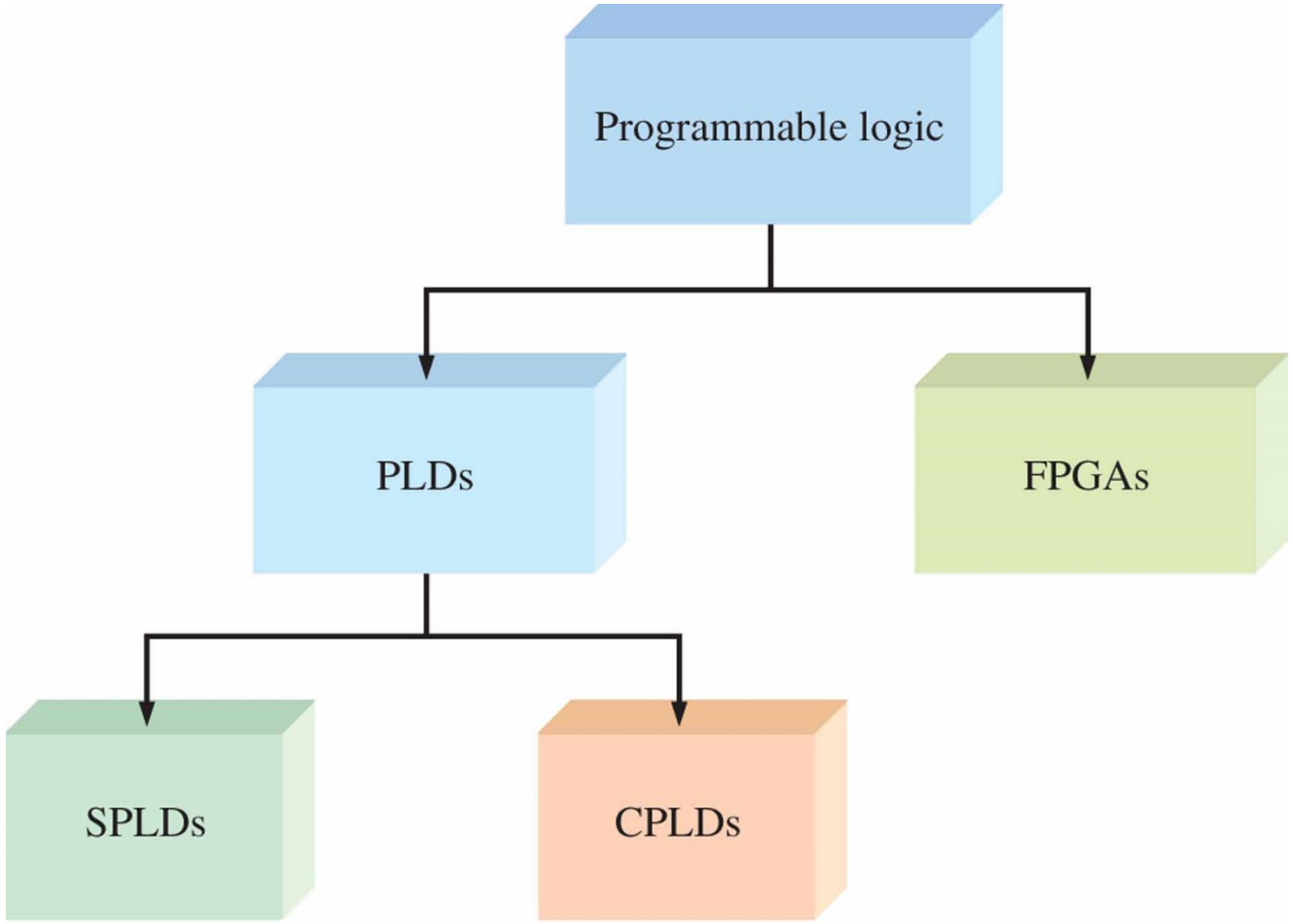
6. Programmable Logic

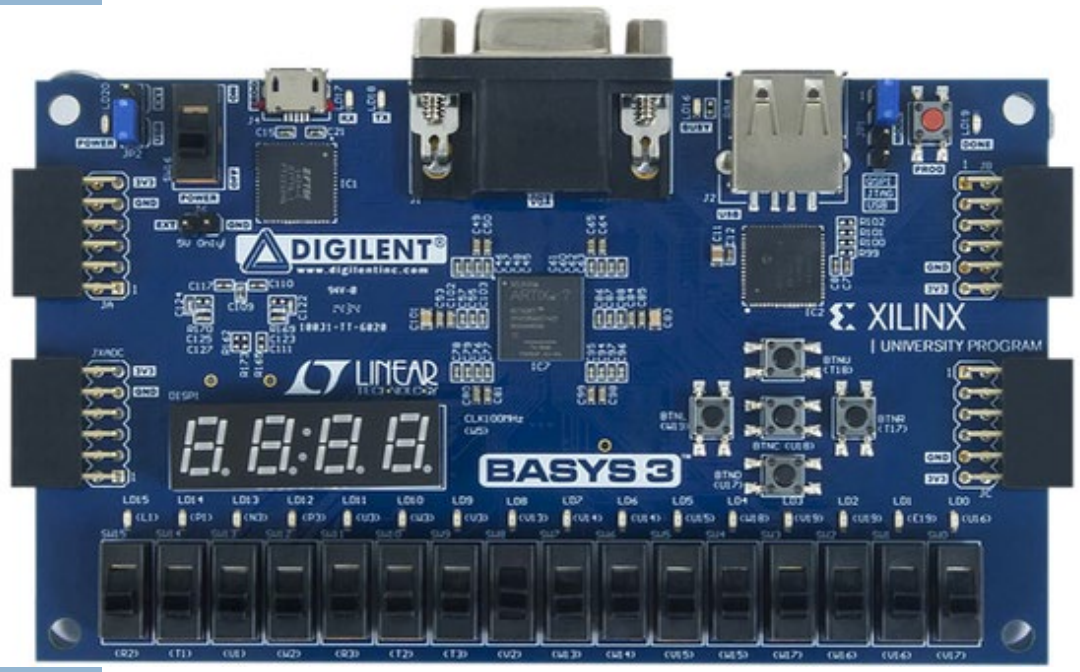
Programmable Logic

- Programmable logic devices can be programmed to perform specified logic functions and operations by the manufacturer or by the user.
- Advantages over Fixed-function logic devices
  - Require less space for an same amount of logic.
  - Change easily without rewiring or replacing components.
  - Implement faster so reduce cost.

Programable Logic Devices



Field-Programmable Gate Array - FPGA



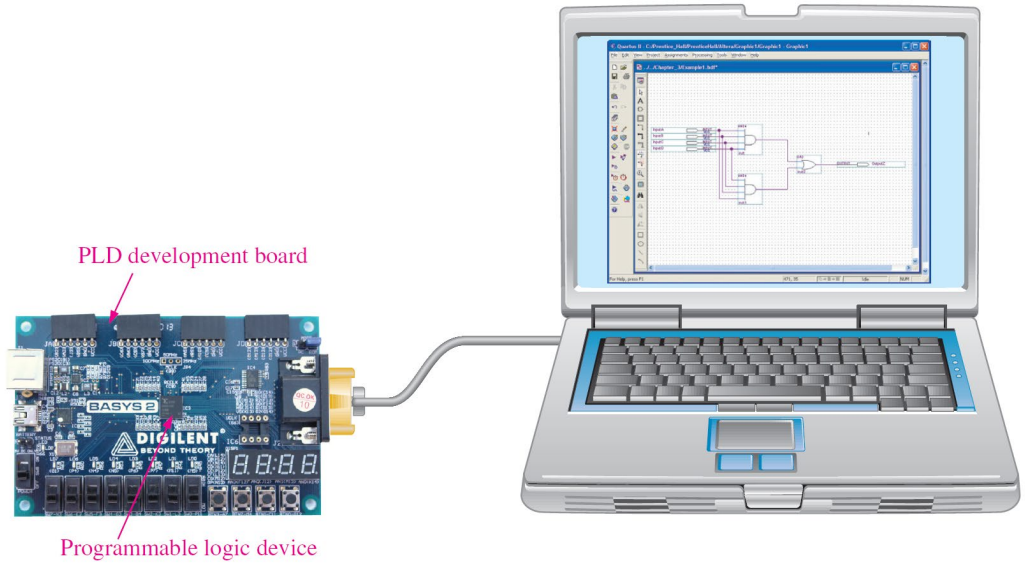


FIGURE 1–36 Basic setup for programming a PLD or FPGA. Graphic entry of a logic circuit is shown for illustration. Text entry such as VHDL can also be used. (Photo courtesy of Digilent, Inc.)

Summary

- Analog is continuous.
- Digital is discreet.
- Binary and Logic represent digital circuit.
- 0-1, True-False, Low-High
- AND, OR, NOT are basic logic functions.

Summary

- Combination Circuit
  - Comparator, Adder, Encoder, Decoder, Mux, DeMux
- Sequential Circuit
  - Flip-flop, Shift register, Counter
- Integrated Circuit : Fixed Logic Device
- Programable Logic Device (PLD)